LESSON 4 • TERM 1 • LEVEL 1

# 100 Square Explorer

Whole numbers: patterns in rows and columns

**TODAY WE WILL**

Explore the 100 square and find the patterns hiding in its rows and columns.



# Get ready

0–4 MIN

- Your 100 square (the kit page)
- About 12 counters, all one kind
- Two or three colours
- Workbook, pencil and paper

**Counters stay on the table, never near the mouth or ears.**

4–10 MIN

# Warm-up: count in tens

Show both hands for every ten

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |

Now count back from 100. Whisper, and shout on 50!




Count along with me!

4–10 MIN

# Fives and twos

One hand is five fingers. Count in fives:

5 10 15 20 25 30

Socks come in pairs. A pair is two. Count in twos:

2 4 6 8 10 12

4–10 MIN

# What comes next?

10 20 30 ___

5 10 15 ___

2 4 6 ___

28 29 ___

38 39 ___

Say the next number. Show it on your fingers if you like.

# Sneak-peek puzzles

10–14 MIN

**Ideas we have not learnt yet. A guess is a great start. Nobody is marked.**

**PUZZLE 1: anyone can try**

Count in 5s. Write the next two numbers.

| 5 | 10 | 15 | 20 | ___ | ___ |
|---|---|---|---|---|---|

**PUZZLE 2: think**

This is a piece of the 100 square. A counter is on 36.

Ravi moves it down one square.

Which number is it on now? How do you know?

| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
|---|---|---|---|---|---|---|---|---|---|
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| | | | | | | | | | |

# Explore: read the square

14–28 MIN

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

1. Put a finger on 1. Slide it along to 10.
2. Where does 11 sit? Slide your finger down.
3. Find 100. Which corner is it in?

# Explore: cover the tens

14–28 MIN

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

1. Count in tens: 10, 20, 30, ...
2. Put a counter on each number you say.
3. What do you see?

SAY IT

The counters make a ___.

# Explore: fives and twos

14–28 MIN

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |

1. Clear your counters. Count in fives to 50.
2. What do you see?
3. Clear again. Count in twos to 20.
4. Which numbers have no counter?

**Will 37 get a counter? Will 60?**

14–28 MIN

# Explore: move the counter

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |

1. Counter on 34. Move it right 1.
2. Now down 1. Down 1 again.
3. Left 1. Then up 1.

WHAT DO YOU NOTICE?

Which way adds 1?

Which way adds 10?

28–38 MIN

# Across: 1 more, 1 less

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |







**Right: 1 more. Left: 1 less.**

At the end of a row, the next number starts the next row: 39, 40, 41.

28–38 MIN

# Down: 10 more, up: 10 less

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |




**Down: 10 more. Up: 10 less.**

A row has ten squares. So one step down is ten more.

28–38 MIN

# Your turn: write the number

Write on your paper. Timer: 1 minute

1

10 more than 36

2

10 less than 58

3

1 more than 29

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

28–38 MIN

# Check: write the number

1

10 more than 36

**46**

2

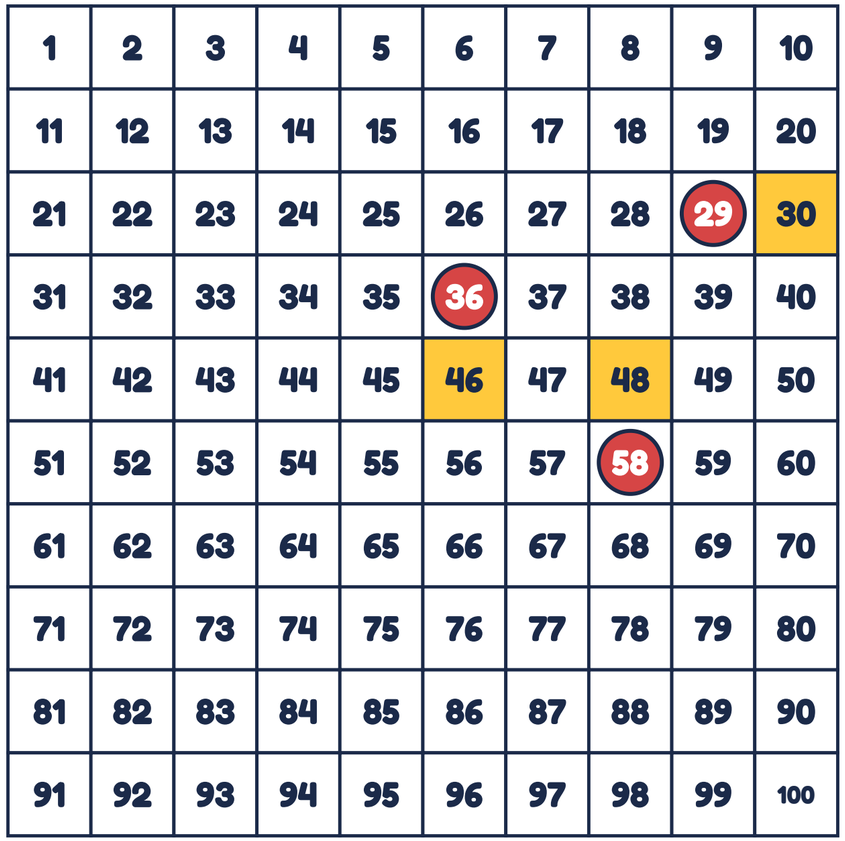
10 less than 58

**48**

3

1 more than 29

**30**



| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

28–38 MIN

# Stripes and last digits

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

## 10s: one column

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

## 5s: two columns

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

## 2s: stripes

THE LAST DIGIT

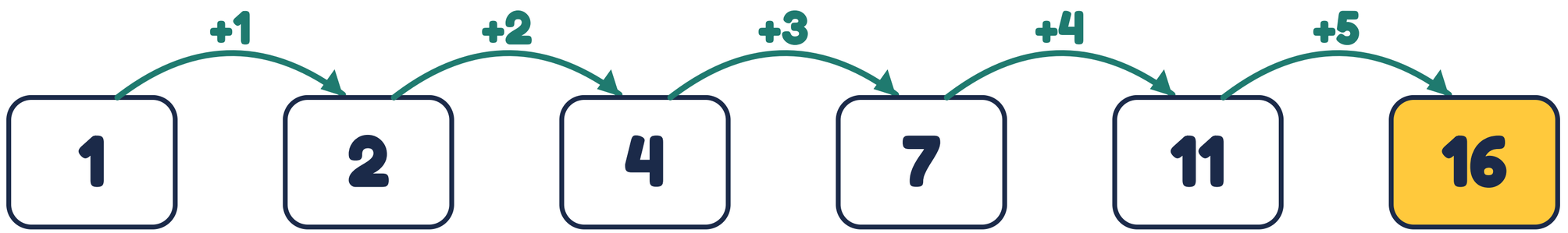
10s end in 0.

5s end in 0 or 5.

2s end in 0, 2, 4, 6 or 8.

28–38 MIN

# Gaps can grow




The gap grows by 1 each time. So the next gap is 5.

28–38 MIN

# Quick-fire: 10 more

**23** 47 **65** 90

Say 10 more. Think: one row down.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

38–42 MIN

# Jump, clap, tap

A big jump for every ten. A clap for every five. A tiny tap for every two.

10 5 2

Sip some water.

42–70 MIN

# Workbook page 1: Start here

Questions 1 and 2. Timer: 8 minutes

**S1** Count in 10s. Colour 10, 20, 30, and keep going to 100. Then fill in the blanks.

**S2** Count in 5s. Colour 5, 10, 15, and keep going to 100. Then fill in the blanks.

42–70 MIN

# Check page 1

| | |
|---|---|
| S1 | They end in 0. They make a line going down. |
| S2 | They end in 5 or 0. They make 2 lines going down. |

42–70 MIN

# Workbook page 2: Start here

Questions 3 to 6. Timer: 10 minutes

**S3** Write the next two numbers in each pattern.

**S4** Count in 2s. Colour 2, 4, 6, and keep going to 30. Write the last digits you see.

**S5** This is a piece of the 100 square. Write the number in each empty square.

**S6** Siti counts in 5s: 5, 10, 15, 20, 26, 30, 35. One number is wrong. Circle it. Write the number she should have said. How do you know?

# Check page 2

42–70 MIN

| | |
|---|---|
| S3 | 10, 12 \| 30, 35 \| 60, 70 \| 41, 51 |
| S4 | 2, 4, 6, 8, 0 (any order) |
| S5 | a 34, b 43, c 55 |
| S6 | 26 is wrong. She should have said 25. Counting in 5s ends in 0 or 5. |

42–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | Our idea in a new setting |
| **Go further (page 4)** | Bigger puzzles |
| **Investigation (page 5)** | Counting in 3s |
| **Check** | Finish and check your own work |

**Everyone may choose. You can swap at any time.**

70–76 MIN

# Talk and explain

ALI SAYS

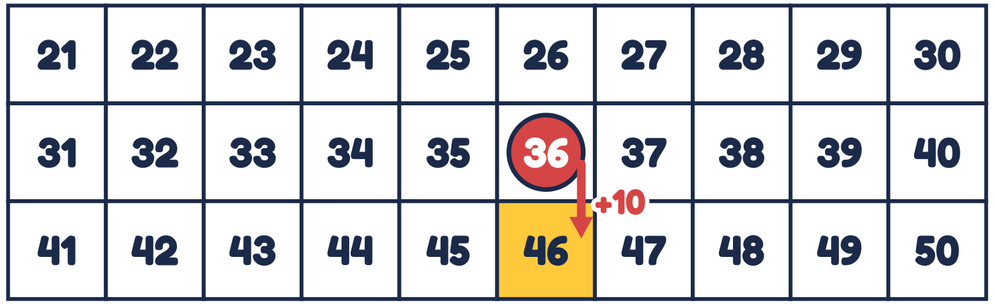
Counting in 5s makes one column on the square, just like counting in 10s.

Do you agree? Use the 100 square to explain.

I think ... because ...

# Back to the sneak-peek

70–76 MIN




**The counter is on 46. Down one row is 10 more.**

76–82 MIN

# Exit ticket

On your own. No looking back. Timer: 3 minutes

**E1** Count in 5s. Write the next two numbers.

**E2** A counter is on 26. It moves down 1, then right 1.

**E3** Mei counts in 2s. She says 25. Is Mei right? Say why.

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | 50, 55 |
| E2 | 36, then 37 |
| E3 | No. 25 ends in 5. Counting in 2s ends in 0, 2, 4, 6 or 8. |

# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ finding my place on the square ☐ counting in 2s, 5s or 10s ☐ moving down a row<br>☐ the pattern where the gap grows (Think harder) ☐ something else |
| **What helped me** | ☐ the 100 square or counters ☐ drawing it ☐ trying again ☐ I asked for help<br>☐ I was not stuck |
| **Next time I will try** | ☐ drawing it ☐ using counters ☐ checking my answer ☐ asking a question |
| | ☐ I am still stuck (my tutor will come and help) ☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

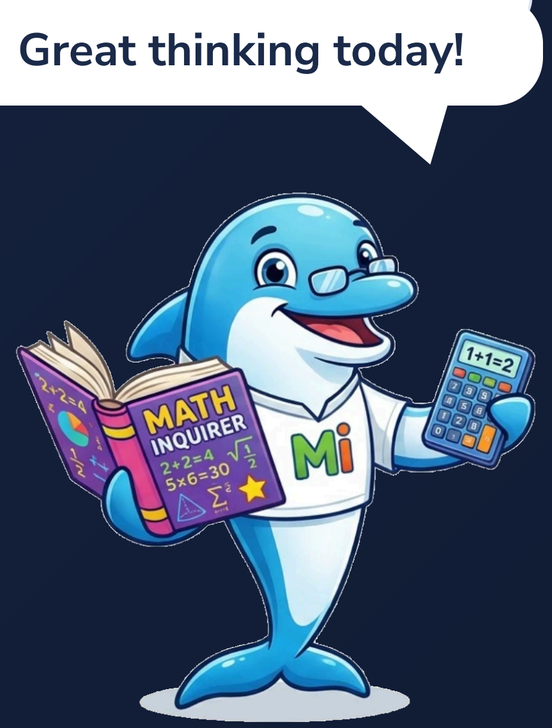
# Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Adding by making tens

You need a pencil and paper.



# Go further: what if?

OPTIONAL

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 |

What if each row had only 5 numbers?

A counter is on 8. It moves down one row. Which number is it on?

What does going down add now?

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | Add 10 to a number. Does only the tens digit change? |
| **Make a puzzle** | Hide a number on the square and write clues for your grown-up |
| **Teach it back** | Show why counting in 10s makes a column |

**Optional. Everyone may join.**